The basics of container stowage planning and why it is so important

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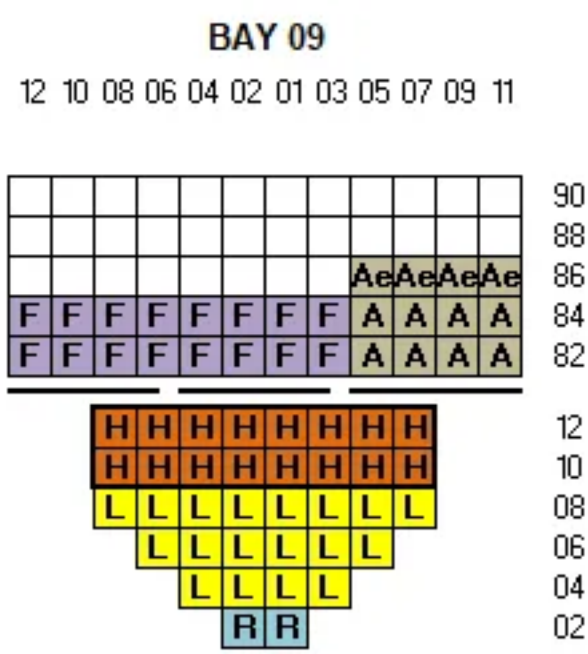


In view of the recent incidents relating to stack collapses on board and containers lost at sea, I thought a refresher on "What is container stowage planning and how it works" would be in order..

This is also to remind everyone in the shipping chain – exporters, agents, freight forwarders, carriers, documentation teams, vessel operators, and stowage planners, the importance of providing and using the correct information for the planning and stowage of the vessel..

So, what is stowage planning..??

In simple terms, it is the act of allocating space on board the ship for containers that have to be loaded from a certain port(s) to be discharged at certain port(s) without those containers having to be rehandled at any of the way ports along the route..

This is probably the most important of all ship operation functions and can be quite intensive in terms of activities and functions..

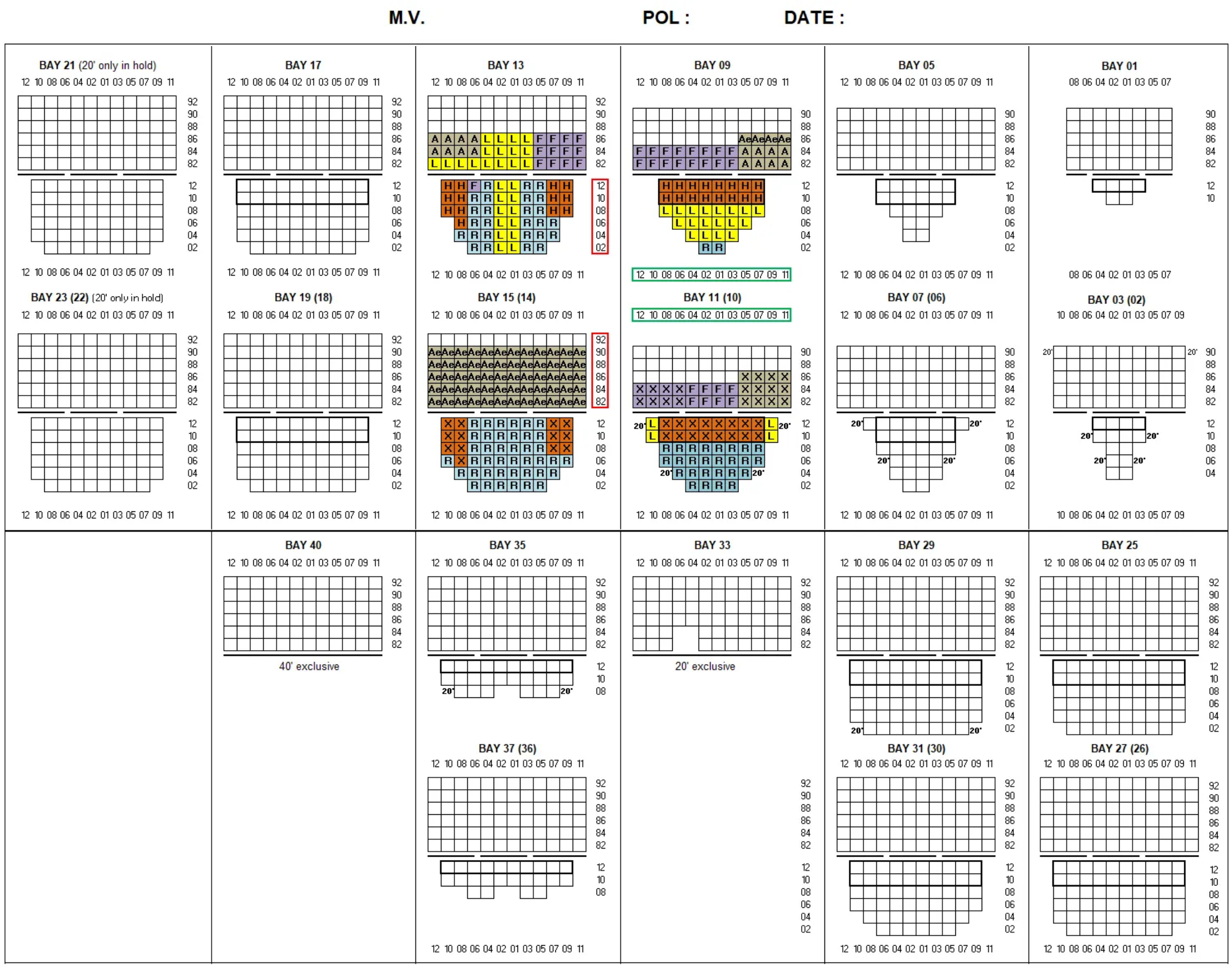
Tools required :

1. The scheduled list of ports that the ship will be calling at, in the order of rotation ;
2. A summary of the number of containers – size/type/weight of containers and container numbers per port that are planned to be loaded on the ship ;
3. A summary of the number of hazardous, reefer and OOG containers per port that are planned to be loaded on the ship ;
4. List and summary of containers on board already on board for ports past your port.. For the purposes of this article, we will consider this port to be Durban..

Definitions :

Profile – is the cross-sectional view of the entire ship covering both the deck and under-deck of the ship.. The profile gives the total view of the stow positions of which containers are to be discharged at which port..

The port operations team and stevedores can identify the sequence of loading and which bays the ship must discharge and/or load from looking at this profile..

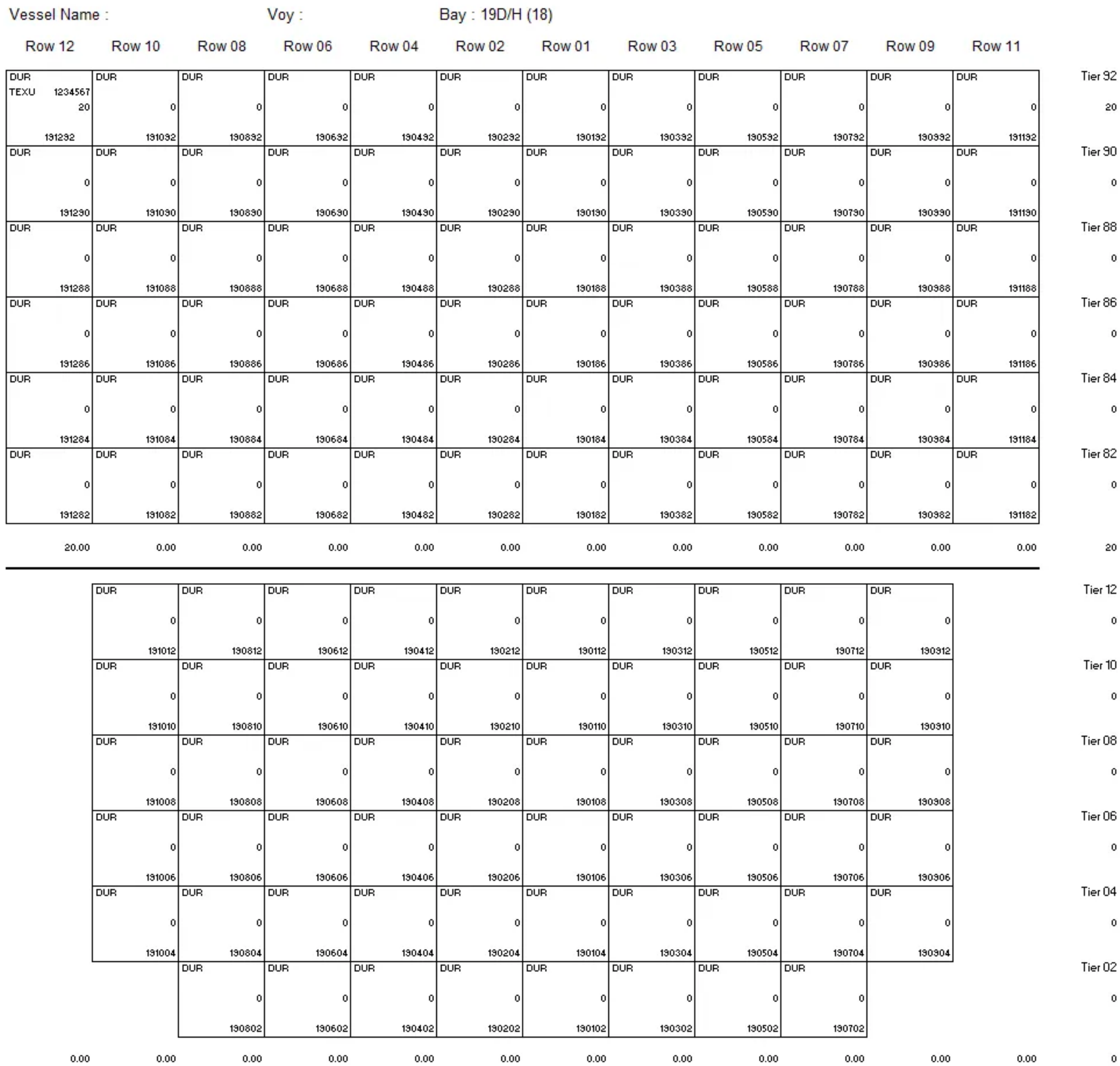


The profile provides the full cross section of a ship at one glance.. The enlarged version of this will be the actual bay itself..

Bayplan – is the complete cross sectional view file covering both the deck and under-deck of the entire ship, but displayed or printed per bay..

Bay – each container vessel is split into compartments which are termed as Bay and depending on the size of the ship it will proceed from 01 to 88 bays (you can read my take on an interesting comparison between stowage plans of older ships and current Triple E type ships) where Bay 01 is the bay towards the Bow (the front) of the ship and Bay 88 is the Stern (the back) of the ship..

Below is the view of Bay 19..



Vessel Name : | Voy : | Bay : 19D/H (18)

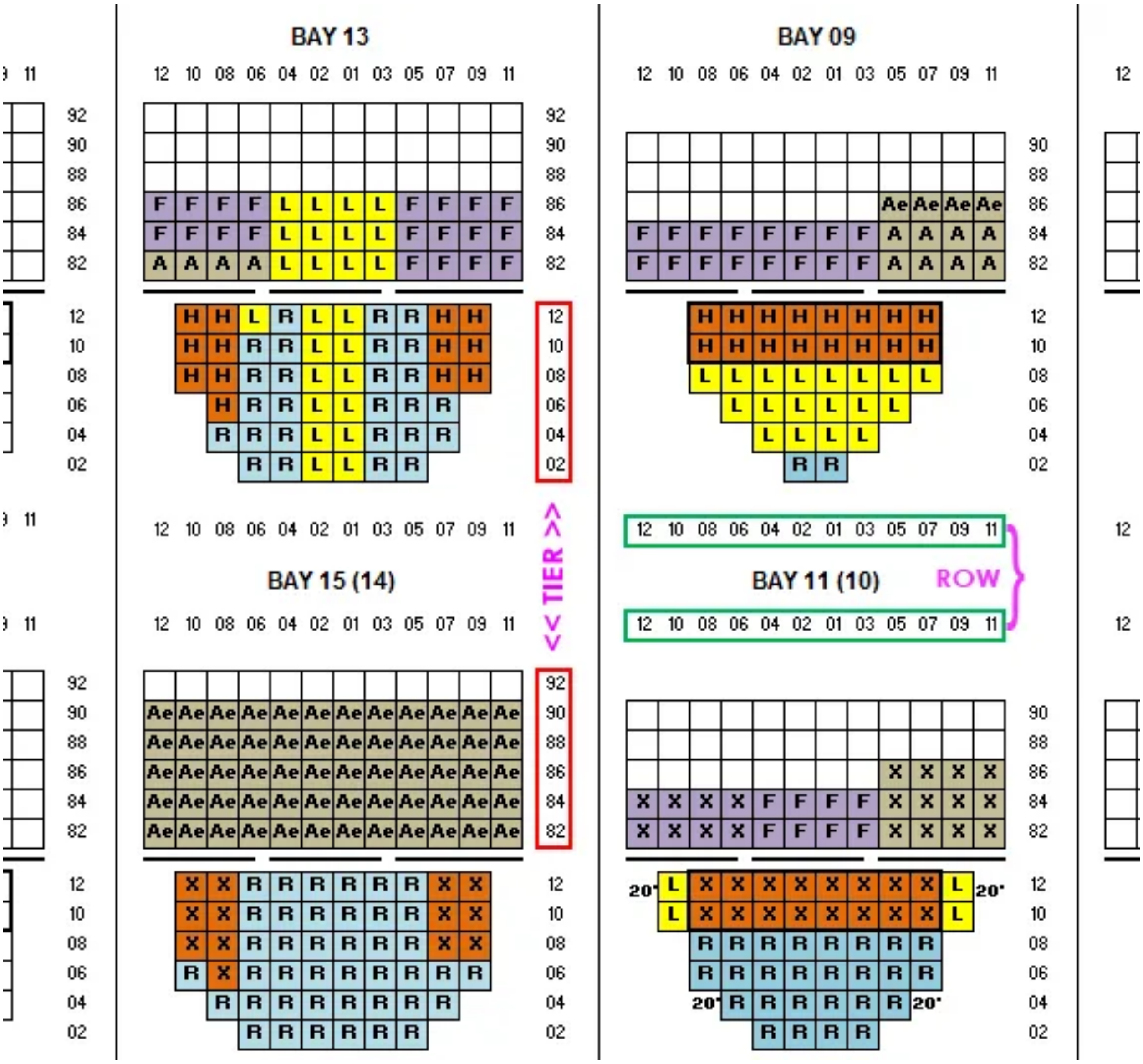
	Row 12	Row 10	Row 08	Row 06	Row 04	Row 02	Row 01	Row 03	Row 05	Row 07	Row 09	Row 11	
Tier 92	DUR TEXU 1234567 20 191292	DUR 0 191092	DUR 0 190892	DUR 0 190692	DUR 0 190492	DUR 0 190292	DUR 0 190192	DUR 0 190392	DUR 0 190592	DUR 0 190792	DUR 0 190992	DUR 0 191192	20
Tier 90	DUR 0 191290	DUR 0 191090	DUR 0 190890	DUR 0 190690	DUR 0 190490	DUR 0 190290	DUR 0 190190	DUR 0 190390	DUR 0 190590	DUR 0 190790	DUR 0 190990	DUR 0 191190	0
Tier 88	DUR 0 191288	DUR 0 191088	DUR 0 190888	DUR 0 190688	DUR 0 190488	DUR 0 190288	DUR 0 190188	DUR 0 190388	DUR 0 190588	DUR 0 190788	DUR 0 190988	DUR 0 191188	0
Tier 86	DUR 0 191286	DUR 0 191086	DUR 0 190886	DUR 0 190686	DUR 0 190486	DUR 0 190286	DUR 0 190186	DUR 0 190386	DUR 0 190586	DUR 0 190786	DUR 0 190986	DUR 0 191186	0
Tier 84	DUR 0 191284	DUR 0 191084	DUR 0 190884	DUR 0 190684	DUR 0 190484	DUR 0 190284	DUR 0 190184	DUR 0 190384	DUR 0 190584	DUR 0 190784	DUR 0 190984	DUR 0 191184	0
Tier 82	DUR 0 191282	DUR 0 191082	DUR 0 190882	DUR 0 190682	DUR 0 190482	DUR 0 190282	DUR 0 190182	DUR 0 190382	DUR 0 190582	DUR 0 190782	DUR 0 190982	DUR 0 191182	0
	20.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20

	Row 12	Row 10	Row 08	Row 06	Row 04	Row 02	Row 01	Row 03	Row 05	Row 07	Row 09	Row 11	
Tier 12		DUR 0 191012	DUR 0 190812	DUR 0 190612	DUR 0 190412	DUR 0 190212	DUR 0 190112	DUR 0 190312	DUR 0 190512	DUR 0 190712	DUR 0 190912		0
Tier 10		DUR 0 191010	DUR 0 190810	DUR 0 190610	DUR 0 190410	DUR 0 190210	DUR 0 190110	DUR 0 190310	DUR 0 190510	DUR 0 190710	DUR 0 190910		0
Tier 08		DUR 0 191008	DUR 0 190808	DUR 0 190608	DUR 0 190408	DUR 0 190208	DUR 0 190108	DUR 0 190308	DUR 0 190508	DUR 0 190708	DUR 0 190908		0
Tier 06		DUR 0 191006	DUR 0 190806	DUR 0 190606	DUR 0 190406	DUR 0 190206	DUR 0 190106	DUR 0 190306	DUR 0 190506	DUR 0 190706	DUR 0 190906		0
Tier 04		DUR 0 191004	DUR 0 190804	DUR 0 190604	DUR 0 190404	DUR 0 190204	DUR 0 190104	DUR 0 190304	DUR 0 190504	DUR 0 190704	DUR 0 190904		0
Tier 02			DUR 0 190802	DUR 0 190602	DUR 0 190402	DUR 0 190202	DUR 0 190102	DUR 0 190302	DUR 0 190502	DUR 0 190702			0
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0

Odd numbered bays (1,3,5 etc) means that it is a 20’ container stow and Even numbered bay (2,4,6 etc) means that it is a 40’ container stow..

Confused..?? Look at the below picture.. I have used Bay 09/11 (10) and Bay 13/15 (14) as an example here..

What you are seeing here is the cross-section of the ship both on deck and under deck.. Each of the small square blocks represents a 20’ unit space..



Row is the position where the container is placed across the width of the ship.. If you refer to the image above, Row numbers are circled in Green..

It starts with 01 in the center and progresses outwards with odd numbers on the right (starboard) and even numbers on the left (port)..

Tier denotes at which level the container is placed – basically how high the container is stacked on board.. In the above image, Tier numbers are circled in Red..

Hatch Covers (the dark intermittent lines in the above picture) are the covers that separate the deck from the under-deck..

The area above the line is called the deck (the area that is visible to us when we look at the ship) and the area below the line is called under-deck (the area that is not visible to us from outside the ship)..

Although currently, computers do most of the work, the basis on which they work is the tried and tested methods that have been followed for many years around..

- the list of containers that are to be loaded onboard is segregated by destination..
- space is allocated to each of the containers
    - firstly in the order of destination – the farthest destination at the bottom and the next port of call right on top
    - secondly in the order of weight – the heaviest boxes at the bottom and lightest at the top

For reasons of lashing and securing containers, a 40' container can sit on top of two 20's, but two 20's cannot sit on top of 40' (unless it is under deck and surrounded by other containers or within cell guides)..

In the above profile, I have used various alphabets and colors..

1. **F** for Felixstowe
2. **A** for Antwerp
3. **Ae** for Antwerp Empty
4. **H** for Hamburg
5. **L** for Le Havre
6. **R** for Rotterdam
7. **X** to indicate that its a 40' contr..

The rotation for this vessel is Felixstowe, Antwerp, Le Havre, Hamburg and Rotterdam..

So because, Felixstowe is the first port of call after Durban, containers bound for Felixstowe are stacked right on top of other containers followed by containers meant for Antwerp, Le Havre, Hamburg and Rotterdam..

Rotterdam will be the last port of call hence it is right at the bottom of the heap..

In this fashion, the entire ship is filled with the containers that are to be loaded at each load port while also taking into account the containers that are ALREADY present on board from the previous ports..

If you notice in the image on the right, there is a container in stow position **110910** (Bay 11, Row 09, Tier 10) – circled in red and marked L for Le Havre.. By reading the bay plan, one can identify a stow position and where a particular container is located..

A restow is a move where a container is off loaded from on board the ship and put back onto the ship either at the same stow position or a different stow position..

This could be due to incorrect stowage of a container – say loaded wrongly for Le Havre instead of Felixstowe or a change of destination was requested at a later stage to now discharge this container in Felixstowe..

In order to reach this container, all the 12 containers meant for Antwerp (A and Ae) has to be "restowed" because Antwerp is the next port after Felixstowe..

Then the hatch cover (the dark line between the deck and under deck) has to be opened to reach under deck..

Then the 1 container to Le Havre (L) in position **110912** must be "restowed" as well and only then the container in position **110910** can be discharged in Felixstowe..

As you can imagine, this involves considerable cost and wastage of time for the ship to restow 12×40′ containers and 1×20′ container to discharge this one container that was stowed incorrectly..

So to avoid these costs and wastage of time, it is imperative that the right destination, correct weight, and hazardous cargo info if any is accurately passed onto the ship..

Each of the bays have deck stress or tier weight which is the maximum allowed weight that each of the tier/row can carry as per the design of the ship..

For example if there are about 4 containers in a tier each weighing 26 tons, it may not be possible to accommodate all 4 in one tier as this might affect stability due to the heavy nature of the cargo..

However, if there are 5 tiers of empty containers as shown in Bay 15, it might be possible to load.. These calculations will be performed by the computer itself and it will show up as errors..

Some of the most commonly used software for ships planning are CASP, MACS3 and Bulko.. These use the BAPLIE file format structured by UNEDIFACT..

Also interesting to note that a lot of Container stowage is done in centralised hubs these days..

As we have seen in the recent articles relating to container collapses onboard ships, correct stowing planning and proper lashing of containers on board is of utmost importance to avoid may maritime disasters..

BAY 09

12	10	08	06	04	02	01	03	05	07	09	11	
												90
												88
								Ae	Ae	Ae	Ae	86
F	F	F	F	F	F	F	F	A	A	A	A	84
F	F	F	F	F	F	F	F	A	A	A	A	82
		H	H	H	H	H	H	H	H			12
		H	H	H	H	H	H	H	H			10
		L	L	L	L	L	L	L	L			08
			L	L	L	L	L	L				06
				L	L	L	L					04
					R	R						02

12 10 08 06 04 02 01 03 05 07 09 11

BAY 11 (10)

12	10	08	06	04	02	01	03	05	07	09	11	
												90
												88
								X	X	X	X	86
X	X	X	X	F	F	F	F	X	X	X	X	84
X	X	X	X	F	F	F	F	X	X	X	X	82
20'	L	X	X	X	X	X	X	X	X	L	20'	12
	L	X	X	X	X	X	X	X	X	L		10
		R	R	R	R	R	R	R	R			08
		R	R	R	R	R	R	R	R			06
		20'	R	R	R	R	R	R	20'			04
				R	R	R	R					02

12 10 08 06 04 02 01 03 05 07 09 11